Module 5
Monitoring and Managing NFS

About This Module

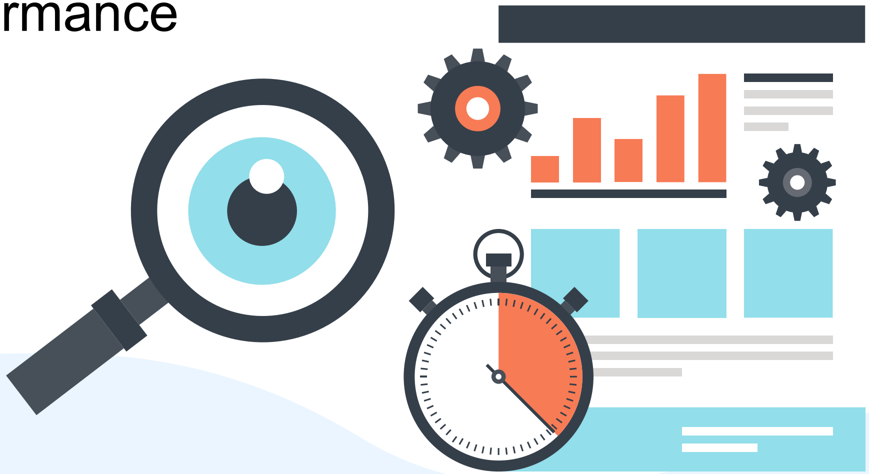
This module focuses on enabling you to do the following:

- Summarize the NetApp monitoring and management applications for NFS
- Illustrate techniques for collecting NFS statistics and data

Lesson 1
Monitoring and Management

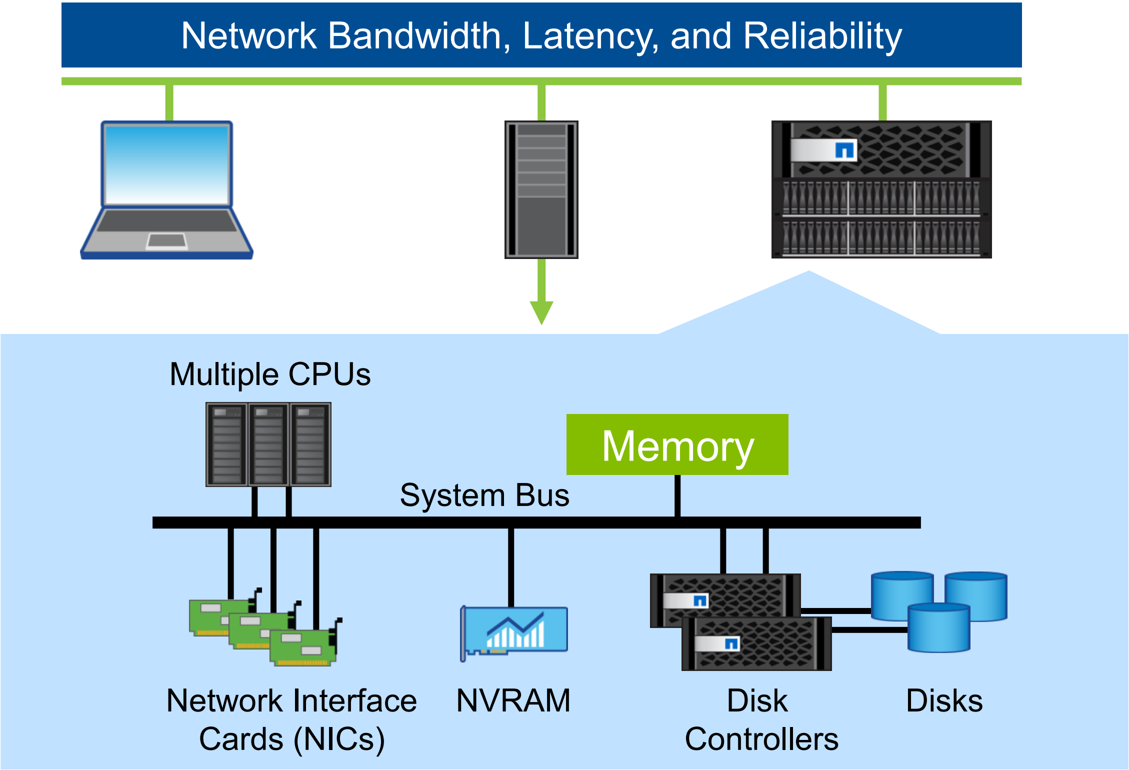
Performance Monitoring Guidelines

- Design and capacity planning
- Establishing baseline system performance
- Monitoring and troubleshooting to maintain established performance
- Optimization to improve system performance



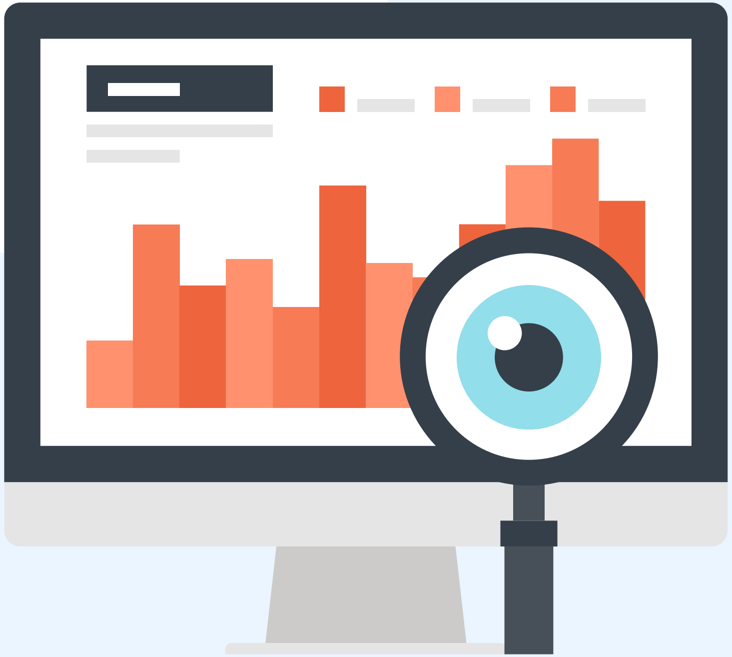
Factors that Affect NFS Performance

- CPU
- Memory
- System bus
- Network
- Network interface
- NVRAM
- I/O devices
  - Disk controllers
  - Disk speeds and RAID



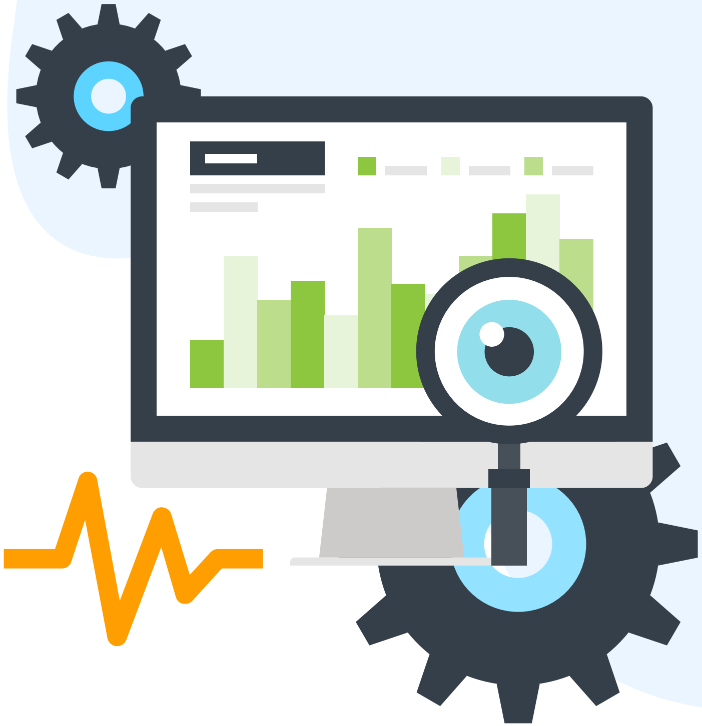
Monitoring and Management Applications

- NetApp ONTAP System Manager
- NetApp Active IQ Unified Manager



ONTAP System Manager

- Configuration of NFS services
  - Enables and disables each version of NFS
  - Configures NFS exports
  - Configures data LIFs
  - Configures name services
- Basic system monitoring and health information



Active IQ Unified Manager

- Real-time performance analytics and event monitoring
- Risk management and remediation
- Capacity management and health monitoring
- Guided troubleshooting and automatic remediation
- Multicluster management
- Active IQ functionality integration
  - Provides a cloud-based intelligence engine
  - Provides predictive analytics and actionable intelligence
  - Simplifies key performance indicators (KPIs) to provide system status and trends that are easy to understand

Lesson 2
Collecting NFS Statistics and Data

Collecting NFS Statistics

- Storage monitoring
  - NFS operations per qtree
  - Statistics by NFS version
  - Packet tracing
  - Per-client statistics
- Client monitoring
  - Load simulation
    - Read and write testing
    - Third-party tools: Iometer and IOzone

Identifying NAS Protocol Traffic

```
cluster1::*>cluster statistics show
          Counter                 Value         Delta
---------------- ----------------- -------------
       CPU Busy:                    4%            -1
        CPU Avg:                    4%            -1
     Operations:
          Total:                     0             -
            NFS:           12957951479     13759s:11s
           SMB:             342195460      230/s:11s
         Fcache:                     0             -
  Total Network:
       Received:                 508MB     292B/s:20s
           Sent:                 602MB     270B/s:20s
     Packets Rx:               2597747        3/s:20s
     Packets Tx:               1738882        2/s:20s
   Data Network:
           Busy:                    0%             -
```

View the values for NFS and SMB. Are they high?

It depends on the site.

Cluster Commands to Monitor NFS Usage

- Use the `statistics catalog object show` command to identify the NFS objects from which you can view data.

```
statistics catalog object show -object nfs*
```

- Examples of selecting counters to monitor: `nfsv3_read_ops` and `nfsv3_write_ops`.

```
cluster1::*>statistics catalog object show -object nfs*

cluster1::*>statistics start -object nfsv3 -instance SVM2 -counter
       nfsv3_read_ops | nfsv3_write_ops -sample-id sample_nfsv3

cluster1::*>statistics stop

cluster1::*>statistics show –sample-id sample_nfsv3
```

NFS and SMB Connections: `statistics show-periodic`

Check performance from the clustershell by using `statistics show-periodic`.

```
cluster1::>statistics show-periodic -node node1
cpu  total                          data  data    data     disk     disk
busy ops    nfs-ops  SMB-ops  busy  recv    sent     read     write
---- -----  -------  -------- ----  ------- -------  -------  --------
54%  10378    10378         0  59%   66.9MB  99.6MB   8.25KB   24.7KB
49%  8156      8156         0  47%   48.0MB  82.0MB   7.92KB   7.92KB
49%  6000      6000         0  54%   24.3MB  87.0MB   15.8KB       0B
56%  10363    10363         0  71%   62.3MB   110MB   8.00KB   24.0KB
```

NFS-ops:	The number of NFS operations per second during that time
SMB-ops:	The number of SMB operations per second during that time
Data busy:	The percentage of time the data network is busy
Data recv:	The amount of data that is received from the data network
Data sent:	The amount of data that is transmitted to the data network
Disk read:	The amount of data that is read from disk (HDD and SSD)
Disk write:	The amount of data that is written to disk

Storage Monitoring: `volume qtree statistics`

ONTAP CLI

```
cluster1::> volume qtree statistics -vserver vsNFS

cluster1::> volume qtree statistics-reset
```

Use this command to zero out statistics.

NFS Statistics: `statistics nfs show`

ONTAP CLI

```
cluster1::>statistics nfs show-v3
cluster1::>statistics nfs show-v4
```

```
cluster1::>statistics nfs show-v4 -node node1
Node               Value   Delta   Percent       Ops     Delta
node1               ---------------------------success------------
Null Procs:        2           -        1%         -
Cmpnd Procs:       92          -                   -
Access Ops:        16          -        6%         -
Close Ops:          8          -        3%         -
Commit Ops:         0          -        0%         -
```

Packet Tracing: `tcpdump`

ONTAP CLI

```
cluster1::>network tcpdump start e0c
cluster1::>network tcpdump stop
cluster1::>network tcpdump trace show
% tcpdump -r /mroot/e0c_<date>_<time>.trc
```

Packet trace files are stored in: /mroot/etc/log/packet_traces.

Identifying the Most Active Client Hosts

Use `statistics top client show` to do the following:

- Display the top 10 most active client hosts
- Sort by total I/O operations

```
cluster1::*>statistics top client show
                                                    *Total              Total
            Client     Vserver               Node      Ops              (Bps)
------------------  ----------  -----------  ------  -----------
192.17.236.53:938   vserver01   cluster1-01      90        20880
192.17.236.160:898  vserver02   cluster1-02      67         5742
```

Identifying the Most Active Files

Use `statistics top file show` to do the following:

- Display the top 10 most active files
- Sort by total I/O operations

```
cluster1::*>statistics top file show
*Total Total
                   File    Volume     Vserver  Aggregate             Node     Ops  (Bps)
----------------------  -------  ---------- ---------- -------------- ------- -----
/vol/vol01/clus/cache     vol01  vserver01       aggr1     cluster1-02       9     80
             /vol/vol02     vol02  vserver02       aggr2     cluster1-01       6     50
```

Resolving Network I/O Bottlenecks

- Distribute the I/O traffic across multiple ports and nodes:
  - Use link aggregation (trunking) to increase network bandwidth.
  - Add more adapters or multiported adapters.
  - Route traffic through under-used interfaces.
  - Upgrade to a 40Gb network interface.

- Distribute I/O to a storage virtual machine (SVM) or volume through different LIFs.

Network Latency

- NFS clients that access storage systems over a high-latency network might see improvement by modifying the TCP maximum read and write size.
- Clients that access storage systems over a low-latency network, such as a LAN, can expect little to no benefit from modifying these parameters.
- Modify the maximum read or write size of NFS transfers.

```
cluster1::*>vserver nfs modify -vserver <svm> -udp-max-xfer-size
<integer>

cluster1::*>vserver nfs modify -vserver <svm> -tcp-max-xfer-size
<integer>
```

- Be sure to increase the read and write I/O sizes on the NFS client hosts: NFS options rsize and wsize.

Pros of Increasing Transfer Size

- **Reduced Overhead:** Moving more data per operation reduces the overall number of Remote Procedure Calls (RPCs), which lowers CPU load on both the client and the storage system.
- **Faster Large-Block I/O:** Favorable for sequential workloads, database full-table scans, and backup/recovery jobs that rely on large data streams.
- **WAN/High-Latency Mitigation:** Helps mitigate the negative impacts of long round-trip times (RTT) for clients connecting over Wide Area Networks (WANs).

Cons of Increasing Transfer Size

- **Higher Memory Consumption:** Large transfers require more memory and buffer space on the ONTAP storage system. Storage systems with low memory might experience performance degradation rather than improvements.
  Increased Latency for Random I/O: Can negatively impact small-block, latency-sensitive workloads because the system has to wait to fill the larger buffers.
- **Network Fragmentation:** If underlying network hardware (like switches or routers) does not support the large packet sizes, fragmentation occurs, which increases latency.
- **Wasted Bandwidth:** If a network error or drop occurs, retransmitting a larger chunk of data takes longer and consumes more bandwidth than smaller chunks.

Monitoring the NFS Mount

ONTAP CLI

```
cluster1::*>vserver nfs modify -vserver <vserver> …
                        -trace-enabled <true/false>
                        -trigger <integer>
```

The SIO Utility

SIO means simulated I/O

- Is a general-purpose load generator
- Enables different block-size read and write operations
- Performs synchronous I/Os to specified files
- Collects basic statistics
- Uses the following syntax:

```
# sio Read% Rand% Blk_Size File_Size Seconds Thread Filename
[Filename]
```

Third-Party Load Generators

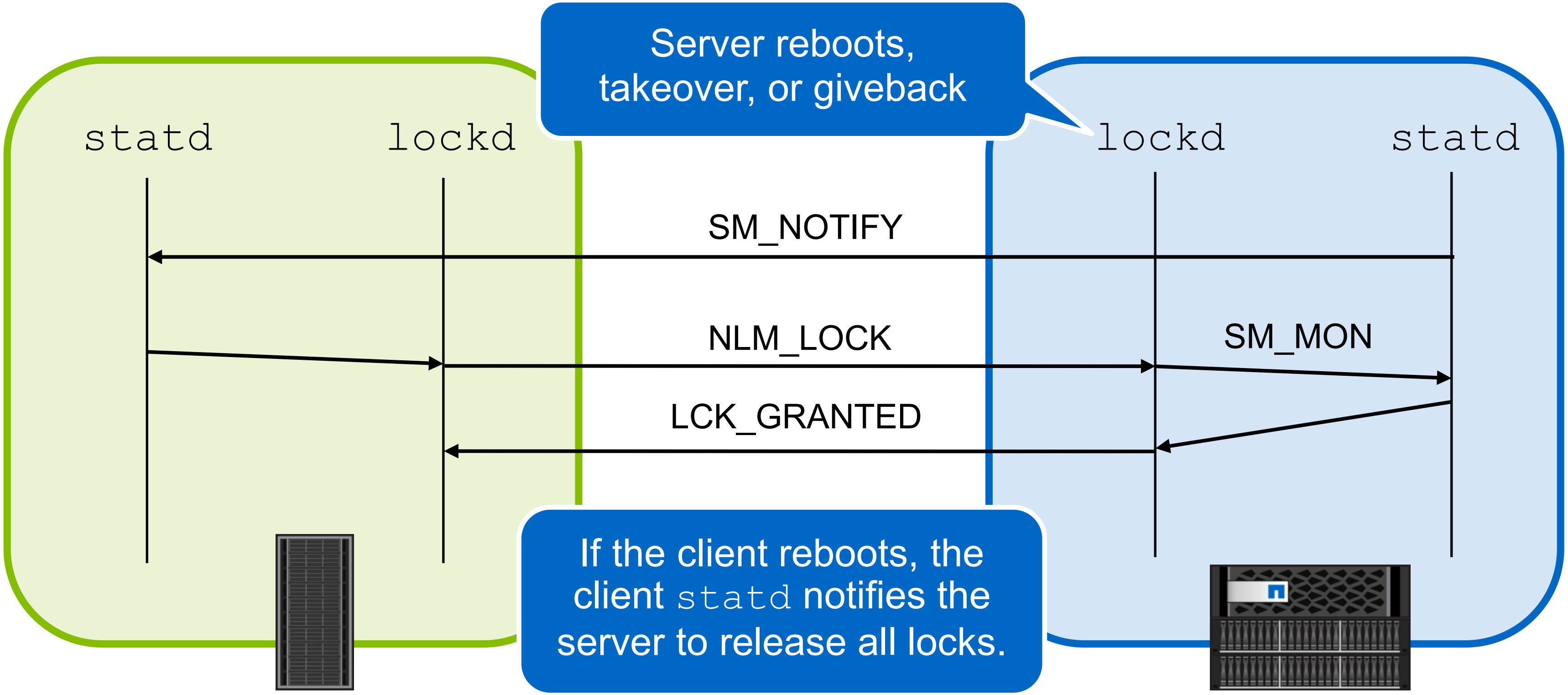
IOzone

- Filesystem benchmark tool
- Generates and measures file operations
- Evaluates I/O performance for various read/write operations
  - Read, read/write, reread, rewrite, read backward, random read/write, and more
- Results can be graphed

Iometer

- Evaluates I/O characteristics
  - Stress tests a system and measures the performance impact
  - Emulates disk or network I/O
- Can measure the following:
  - Bandwidth and latency capabilities of I/O buses
  - Network throughput to attached disks
- Consists of two programs:
  - Iometer: the controlling application
  - Dynamo: the multithreaded workload generator

NFS Lock Recovery Process



Viewing NFSv3 File Locks

ONTAP CLI

```
cluster1::> vserver locks show
```

Viewing an SVM lock is often a first step in determining why a client cannot access a volume or file.

NOTE: Be careful when breaking a lock. Applications can become unstable if lock-breaking is performed incorrectly. When in doubt, contact NetApp Support.

Breaking the NFSv3 File Lock

1. Linux client:
   - Shut down all processes that use the affected NFS resources.
   - Unmount the NFS resources.
   - Stop `statd` and `lockd`.
2. ONTAP CLI:

```
cluster1::>vserver locks show
cluster1::>vserver locks break
   -vserver vserver
   -volume volume
   -path path
   -lif lif
```

3. Linux client:
   - Restart `statd` and `lockd`.
   - Mount the NFS resources.

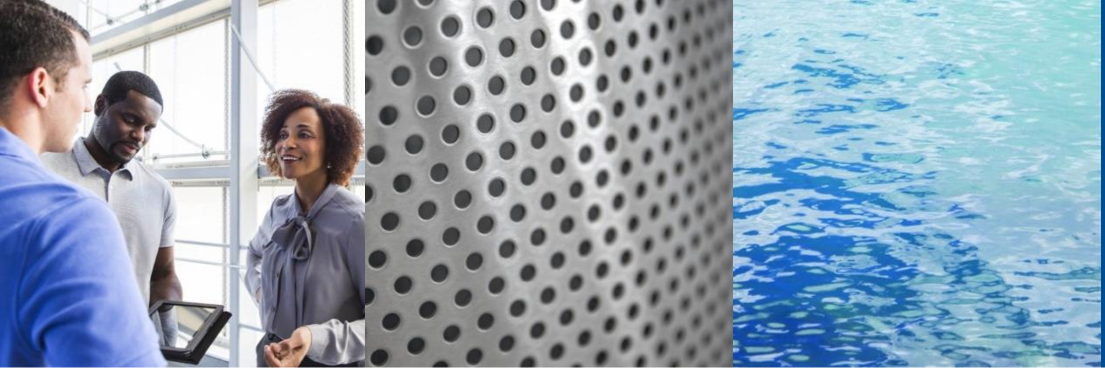
Contact NetApp Support
for assistance.

NFSv4 and NFSv4.1 Lease Locks

ONTAP CLI

```
cluster1::>vserver locks show
```

Because of the way that NFSv4 and later leases work, breaking the lease is rarely required.



Take a Poll

Two-Question Quiz

Duration: 10 minutes

Poll Questions

1. Which of the following tools is used to view a packet trace?
    a. IOzone
    b. Iometer
    c. dd
    d. SIO
    e. Wireshark

Poll Questions: Answers

1. Which of the following tools is used to view a packet trace?
   a. IOzone
   b. Iometer
   c. dd
   d. SIO
   e. Wireshark

Poll Questions

2. Which type of statistics does the `statistics nfs show-v4` command display?
   a. packet traces
   b. statistics for NFSv4 operations and counters
   c. statistics for NFSv4 parallel NFS (pNFS) only
   d. statistics for both NFSv4 and NFSv3 operations and counters

Poll Questions: Answers

2. Which type of statistics does the `statistics nfs show-v4` command display?

   a. packet traces

   b. statistics for NFSv4 operations and counters ✔

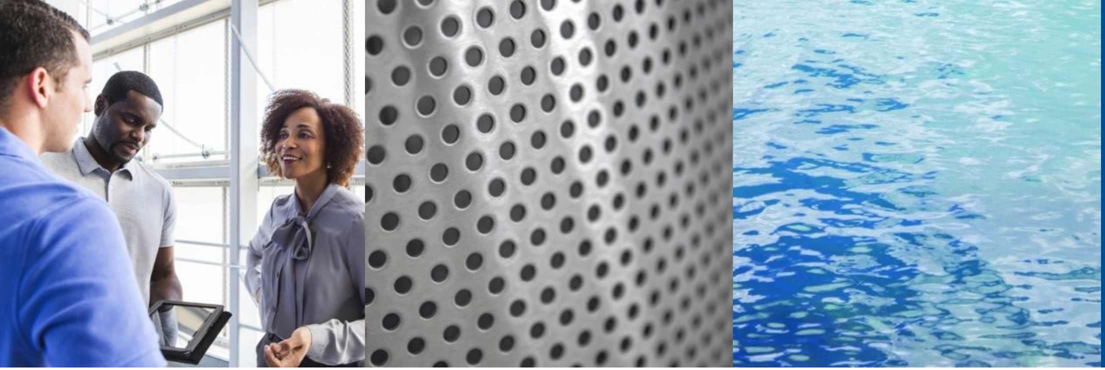
   c. statistics for NFSv4 parallel NFS (pNFS) only

   d. statistics for both NFSv4 and NFSv3 operations and counters

Course Summary

This course focused on enabling you to do the following:

- Explain the differences between the NFS protocol and SAN protocols
- List NAS namespace architectures and namespaces
- Summarize client access to NFS data in ONTAP software
- List the NFS enhancements for different releases of ONTAP software
- Describe the implementation of NFSv3 on NetApp storage
- Describe the implementation of NFSv4 on NetApp storage
- Demonstrate a multiprotocol configuration that supports SMB and NFS access
- Illustrate techniques for collecting NFS statistics and data



Complete an Exercise

Module 5: Monitoring NFS Performance

Duration: 20 minutes

ACTION: Share Your Experiences

Roundtable questions for the equipment-based exercise

What commands did you use to monitor NFS on the node?

Module Summary

This module focused on enabling you to do the following:

- Summarize the NetApp monitoring and management applications for NFS
- Illustrate techniques for collecting NFS statistics and data

YOU ARE HERE

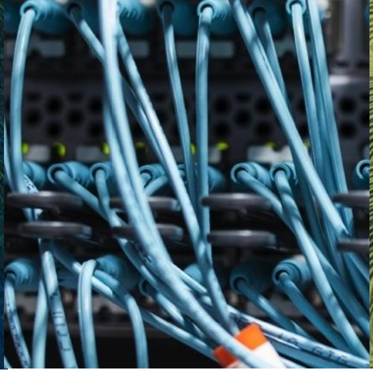
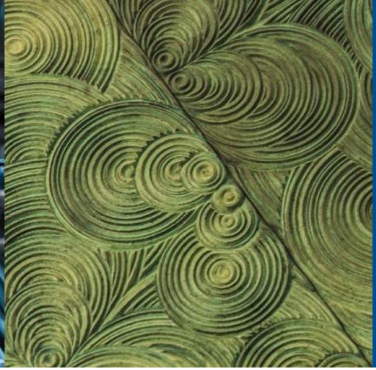
- ✓NFS Fundamentals
- ✓NFS Version 3
- ✓NFS Versions 4 and 4.1
- ✓Multiprotocol Configuration
- ✓Monitoring and Managing NFS

References

- NetApp University courses:
  ONTAP Cluster Administration
- Documentation:
  - ONTAP 9 NFS Reference
  - ONTAP 9 NFS Configuration Power Guide
- Technical Reports:
  - NFS Best Practice and Implementation Guide
  - NetApp ONTAP FlexGroup Volumes: Best Practices and Implementation Guide
  - Name Services Best Practices Guide
  - Secure Unified Authentication: Kerberos, NFSv4, and LDAP in ONTAP
- Community:
  http://communities.netapp.com/community/products_and_solutions

Your Feedback Needed

Please take a few minutes to complete the survey for this course.

Closing Thoughts

Thank You